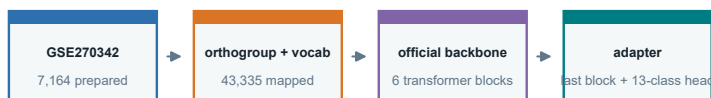
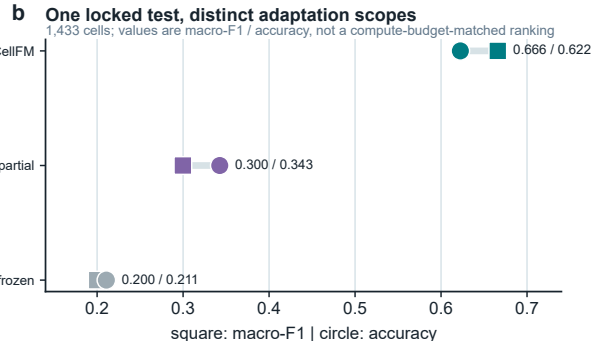


a Matched scPlantLLM references now include partial adaptation

Same GSE270342 object, first-target mapping and exact Plant-CellFM locked test barcodes

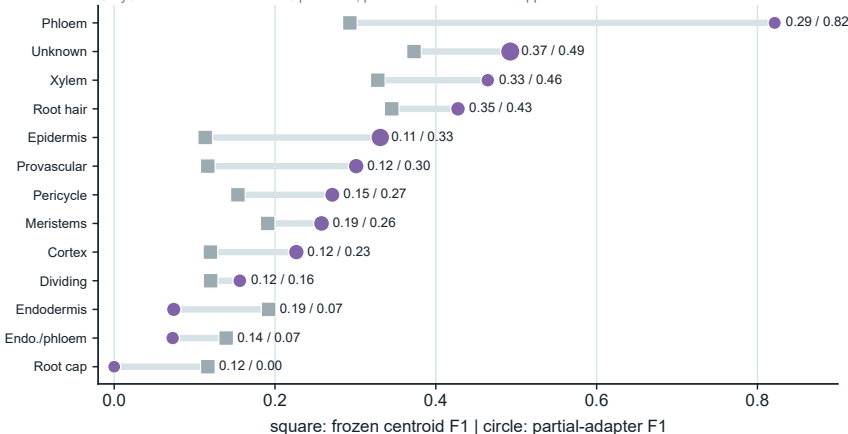


Partial adaptation leaves the first five transformer blocks frozen; best epoch is selected only by validation macro-F1.



c The partial adapter changes per-class behaviour visibly

Grey / violet labels show frozen / partial F1; point area is locked-test support



d Comparator boundary

A matched adaptation reference, not a universal ranking

